

BAYESIAN TESTS FOR HYPOTHESIS OF THE EQUALITY OF MULTINOMIAL PROBABILITIES

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Summary:

A Bayesian test is derived for testing the null hypothesis that the parameters of a multinomial distribution are equal against a composite alternative. A class of symmetric Dirichlet priors is assumed and it appears that a unique lower bound exists for the posterior of the null hypothesis for this class of priors. In the binomial case it is shown that there is little difference between a point null hypothesis and a composite null, or "fuzzy" point null, hypothesis when the interval is moderately small. The Bayes decision based on expected posterior loss is also discussed.

1. Introduction

In this paper we examine a Bayesian test for the equality of multinomial parameters. By Bayesian test is meant the evaluation of the posterior probabilities for the respective hypotheses. The purpose is twofold. Firstly to show that the information obtained from a Bayesian analysis is clearer and more understandable to a non-statistician than a classical test of hypothesis, since the standard question by the non-statistical experimenter

is: "What is the probability that this hypothesis is true?", which is a Bayesian interpretation of the problem. Secondly, if a decision is to be made based on the results of the analysis, a decision-theoretic approach is appropriate.

In section 2 the posterior distributions under the null and alternative hypotheses are derived for a certain class of priors and in section 3 the lower bound on the posterior probability for the null hypothesis is discussed. In section 4 the relationship between a point null and composite null hypothesis is examined in the binomial case. Bayes decision theory is discussed in section 5.

2. Prior and posterior probabilities

Consider a multinomial situation with k possible outcomes with corresponding probabilities $\theta_1, \theta_2, \dots, \theta_k$. We are interested in the hypothesis that all outcomes are equally likely, so let

$$H_0: \theta_i = \frac{1}{k}, i=1, \dots, k$$

against the composite alternative

$$H_1: H_0 \text{ is not true.}$$

So we can divide the parameter space into two parts, namely

$$\Theta = \{\Theta_0, \Theta_1\},$$

where Θ_0 is the single point where $\theta_i = \frac{1}{k}, i=1, \dots, k$ and Θ_1 is the rest of the parameter space,

$$\Theta_1 = \left\{ (\theta_1, \dots, \theta_k): 0 \leq \theta_i \leq 1, \sum \theta_i = 1, (\theta_1, \dots, \theta_k) \neq \left(\frac{1}{k}, \dots, \frac{1}{k} \right) \right\}.$$

The probability model when a sample of n observations is drawn is then

$$(2.1) \quad f(\mathbf{x}|\theta) = \frac{n!}{x_1! \dots x_k!} \prod_{i=1}^k \pi(\theta_i)^{x_i}, \quad 0 \leq x_i \leq n, \\ \sum x_i = n$$

which reduces to

$$(2.2) \quad f(\mathbf{x}|\theta) = \frac{n!}{x_1! \dots x_k!} \left(\frac{1}{k} \right)^n, \quad 0 \leq x_i \leq n \\ \sum x_i = n$$

under the null hypothesis.

A prior distribution must be placed on the parameter space $\bar{\Theta}$. This prior should be proper and should place a positive mass at the point $\bar{\Theta}_0$ while distributing the rest of the mass over $\bar{\Theta}_1$.

Now the mass, say p , that is placed on $\bar{\Theta}_0$ is subjective and would depend on the experimenter's assessment in a given situation. Usually, to stay impartial with regard to the two hypotheses, a "neutral" value of $p = \frac{1}{2}$ can be assumed. However, the Bayes factor (Berger (1985) p. 146) is a convenient measure by which to express posterior confidence in H_0 and avoids specification of the prior probability p . To be noninformative under the alternative, the rest of the prior mass can be distributed uniformly over $\bar{\Theta}_1$. More generally, the class of symmetric Dirichlet distributions with parameter α is assumed under H_1 where $\alpha \geq -\frac{1}{2}$. This is a logical class of priors since it does not favour any particular one of the θ_i 's and includes the uniform case. The reason for the lower bound on α is to include Jeffreys' prior in the class (Jeffreys (1961)). If the experimenter suspects that the θ_i 's will be close to each other when they are not equal, a large value of α is appropriate, while values close to zero will define a relatively noninformative prior. For suggestions of a Type III distribution on α , see Good (1967). Good also shows by means of examples that no single Dirichlet prior with fixed α is adequate for all cases of significance testing.

So the prior is defined as

$$(2.3) \quad \pi(\theta) = \begin{cases} p & \theta \in \bar{\Theta}_0 \\ (1-p) \frac{\Gamma[k(\alpha+1)]}{(\Gamma[\alpha+1])^k} \frac{k}{1} \pi_1(\theta_i^\alpha), & \theta \in \bar{\Theta}_1 \end{cases}$$

and the posterior according to Bayes' theorem is (see DeGroot (1958) p. 238, Berger (1985))

$$\pi(\theta|x) = \begin{cases} \pi_0(\theta|x), & \theta \in \bar{\Theta}_0 \\ \pi_1(\theta|x) & \theta \in \bar{\Theta}_1 \end{cases}$$

where

$$(2.4) \quad \begin{aligned} \pi_0(\theta|x) &= P\left[\theta = \frac{1}{k} e | x\right] \\ &= \frac{pf(x|\theta = \frac{1}{k} e)}{pf(x|\theta = \frac{1}{k} e) + (1-p) \int_{\bar{\Theta}_1} f(x|\theta) \pi_1(\theta) d\theta} \\ &= \left[1 + \frac{1-p}{p} \frac{k^n \Gamma[k(\alpha+1)] \frac{k}{1} \Gamma[\alpha+x_i+1]}{\Gamma[k(\alpha+1)+n] (\Gamma[\alpha+1])^k} \right]^{-1} \end{aligned}$$

and

$$(2.5) \quad \pi_1(\theta|x) = \pi_0(\theta|x) \frac{1-p}{p} \frac{k^n \Gamma[k(\alpha+1)]}{(\Gamma[\alpha+1])^k} \frac{k}{1} \theta_i^{x_i+\alpha}, \quad \theta \in \bar{\Theta}_1.$$

The Bayes factor in favour of H_0 (Berger (1985) p. 146) is given by

$$(2.6) \quad B(\alpha) = \frac{\Gamma[k(\alpha+1)+n] (\Gamma[\alpha+1])^k}{k^n \Gamma[k(\alpha+1)] \frac{k}{1} \Gamma[\alpha+x_i+1]}.$$

If you are impartial towards the two hypotheses, and in the absence of a loss function, a criterion that makes sense is to determine whether the data support the null hypothesis or not. If it is the case, accept the null hypothesis. So a Bayesian test would be to reject the null hypothesis if

$$\pi_{\theta}(\theta|\mathbf{x}) < \pi_{\theta}(\theta),$$

where $\pi_{\theta}(\theta)$ is the prior probability for H_{θ}

This condition reduces to

$$(2.7) \quad B(\alpha) < 1$$

or

$$(2.8) \quad \prod_{i=1}^k \pi \Gamma[\alpha + x_i + 1] > \frac{\Gamma[k(\alpha + 1) + n] (\Gamma[\alpha + 1])^k}{k^n \Gamma[k(\alpha + 1)]}.$$

3. Lower bound for $\pi_{\theta}(\theta|\mathbf{x})$

One can always find an absolute lower bound on $\pi_{\theta}(\theta|\mathbf{x})$ by choosing the prior under the alternative to be a point mass at the maximum likelihood estimator of θ , say $\hat{\theta}$.

This is however very restrictive since it means putting all the mass at the point most favourable for H_1 , but it does give an absolute lower bound.

In multivariate problems this lower bound is too small to be useful. One possible generalization of this (Berger (1986)) would be to take $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_k)$ and consider all permutations of it,

$$\hat{\theta}^{(i)} = (\hat{\theta}_{i_1}, \hat{\theta}_{i_2}, \dots, \hat{\theta}_{i_k}),$$

and then choose $\pi(\theta)$ to be

$$(3.1) \quad \pi(\theta) = \begin{cases} p & , & \theta_i = \frac{1}{k}, \quad i=1, \dots, k. \\ (1-p) \frac{1}{k!} & , & \theta = \hat{\theta}^{(i)}, \quad i=1, \dots, k. \end{cases}$$

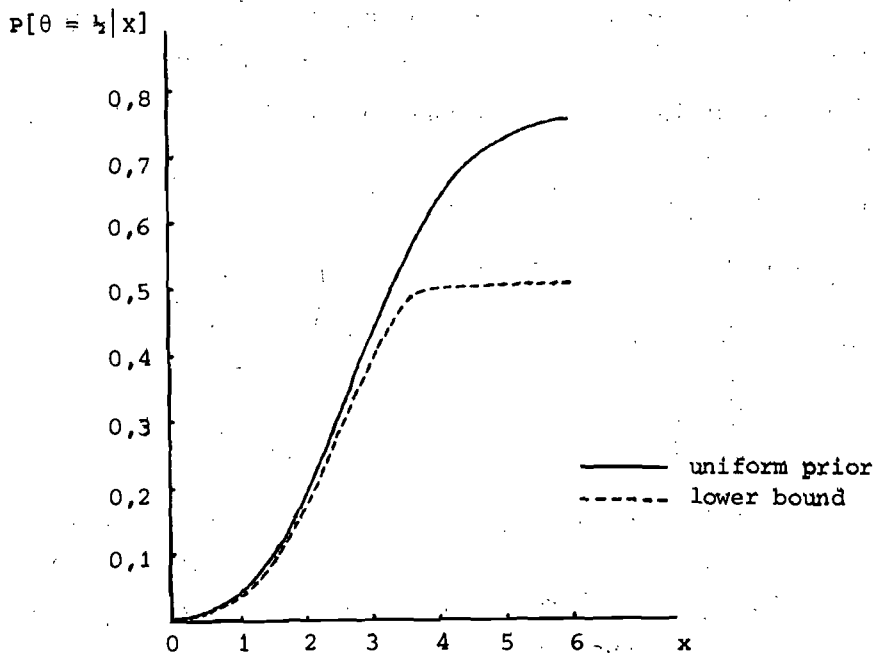
This is then an exchangeable prior with some, but not all, of the mass under the alternative at $\hat{\theta}$. For the class of symmetric Dirichlet priors defined in (2.3) a lower bound exists, but has not yet been determined explicitly. It is at the value of α that minimizes $B(\alpha)$ in (2.6) for $\alpha \geq -\frac{1}{2}$, and can be found by successive enumeration.

In table 1 and figure 1 the lower bound is given in the Binomial case

together with the posterior probability for H_0 when $n = 12$, $p = \frac{1}{2}$ and $x = 1, 2, \dots, \frac{n}{2}$. The values of α at which the lower bound occurs are also given, as well as the P -values for the classical test of the hypothesis $H_0: \theta = \frac{1}{2}$.

Table 1.

x	Uniform prior $\pi_0(\theta x)$	Lower bound $\pi_0(\theta x)$	α	P -value
0	0,0032	0,0015	-0,50	0,0005
1	0,0367	0,0335	-0,42	0,0059
2	0,1732	0,1724	0,17	0,0386
3	0,4112	0,3763	1,70	0,1460
4	0,6111	0,4937	15,50	0,2417
5	0,7154	0,5000	∞	0,7744
6	0,7457	0,5000	∞	1,0000

Figure 1. Posterior probabilities, $P[\theta = \frac{1}{2} | x]$ for $n = 12$.

Note that when the data support the null hypothesis, the lower bound for the probability that H_0 is true is attained when the mass under H_1 is more concentrated around the point $\theta = \frac{1}{2}$, as reflected by a large value of α . In fact, when x is close to $\frac{n}{2}$ the lower bound of 0.5 is approached only when $\alpha \rightarrow \infty$.

At the other extreme, when $x = 0$ or n , the posterior probability for H_0 under the uniform prior ($\alpha=0$) is given by

$$(3.2) \quad P[\theta=\frac{1}{2} | x=0 \text{ or } n] = \left[1 + \frac{2^n}{n+1} \right]^{-1}$$

while the lower bound is attained at the lower bound we set for α , namely $\alpha = -\frac{1}{2}$.

$$(3.3) \quad \inf_{\alpha \geq -\frac{1}{2}} P[\theta=\frac{1}{2} | x=0 \text{ or } n] = \left[1 + \frac{\frac{n}{2} \pi(2i-1)}{n!} \right]^{-1}$$

From table 1 it can also be seen that the P -values in the tail are considerably lower than the lower bound on the posterior probabilities. This is a phenomenon present in all Bayesian hypothesis testing (see for example, Pratt (1965), Casella & Berger (1987), Dickey (1977)) and shows again that it can be misleading to give a Bayesian interpretation to a classical P -value.

Example

In a recent paper by Meyer *et al.* (1985) the results were given of an experiment to compare bio-availability of four formulations of a certain drug. We will refer to the formulations as A , F , L and P . Twelve volunteers participated in the study and the success of a drug is measured in terms of the AUC values (the area under the plasma concentration time curve). Meyer, *et al.* (1985) concluded that in terms of the above-mentioned value, brand L is superior to the other three, using a three factor analysis of variance.

A non-parametric Bayesian analysis of the data (which are not reproduced here) will be done. We are not suggesting that bio-equivalence

should be solely based on this analysis, since a certain amount of information is discarded, but it can serve as a useful additional tool for assessment, especially if there exists some doubt about the distributional assumptions with regard to the data. Also, the use of the AUC values in this example does not imply that it is the only (or best) measure for bio-equivalence. Meyer *et al.* (1985) actually used five different measures for comparing the four formulations. The data will be treated as multinomial with a success recorded for the drug with the highest AUC value. Let X_A , X_F , X_L and X_P be the respective variables, then for the $n=12$ observations the result was

$$(x_A, x_F, x_L, x_P) = (2, 1, 7, 2).$$

Let θ_A , θ_F , θ_L and θ_P be the respective probabilities for a formulation to give the highest AUC value, then we want to test

$$H_0: \theta_i = \frac{1}{4}, \quad i=A, F, L, P$$

against

$$H_1: H_0 \text{ not true.}$$

The probabilities for the outcome $(2, 1, 7, 2)$ under H_0 is 0,0014. The P -value has been calculated by adding up all the multinomial probabilities for $n=12$ from the smallest until 0,0014 at observation $(2, 1, 7, 2)$ and all its permutations has been reached. This yields $P = 0,1102$ which means that H_0 cannot be rejected. With the Bayesian analysis it follows from (2.4) with $p=\frac{1}{4}$ and uniform prior ($\alpha=0$) that the posterior probability for H_0 is given by

$$(3.4) \quad P\left[\theta = \frac{1}{4} \mathbf{e} | \mathbf{x} = (2, 1, 7, 2)\right] = 0,3918.$$

This suggests mild evidence against H_0 . The Bayes factor is $B(0) = 0,64$ and can be interpreted as the comparative evidence provided by the data for H_0 against H_1 .

Thus according to the criterion (2.7), the null hypothesis should be rejected.

The posterior under the alternative is given by (from 2.5)

$$(3.5) \quad \pi_I(\theta | \mathbf{x}) = \frac{6\pi \theta_i^{x_i}}{\left(\frac{1}{4}\right)^n + \frac{6\pi \Gamma(x_i+1)}{\Gamma(n+4)}} \cdot \frac{0 < \theta_i < 1, i=A, F, L, P}{\sum \theta_i = 1}$$

The posterior probabilities under the alternative hypothesis are very informative and can be obtained from (3.5) through numerical integration. What is of interest in this case is the probability that a particular formulation is better than the other three. (For the purpose of this analysis the terms "better" or "best" simply imply the highest probability for the largest AUC value.) Because of the uniform prior on the alternative hypothesis space and with $p=\frac{1}{4}$, the prior probability that any particular formulation is better than the rest is 0,125. Now the posterior probabilities follow from (3.5) as (see appendix A)

$$(3.6) \quad \begin{aligned} P[A \text{ is best} | \mathbf{x} = (2, 1, 7, 2)] &= 0,0290 \\ P[F \text{ is best} | \mathbf{x} = (2, 1, 7, 2)] &= 0,0089 \\ P[L \text{ is best} | \mathbf{x} = (2, 1, 7, 2)] &= 0,5412 \\ P[P \text{ is best} | \mathbf{x} = (2, 1, 7, 2)] &= 0,0290. \end{aligned}$$

Together with (3.4) it follows, for example, that there is a probability of 0,933 that L has an AUC value at least as high, or higher than any other formulation. This kind of posterior probability is quite robust with respect to changes in p , the prior for H_0 , which is not the case for the probabilities under either H_0 or H_1 alone.

Returning to the class of Dirichlet priors, examination of $B(\alpha)$ in (2.6) reveals a unique global maximum at $\alpha=1,51$ which means a lower bound of 0,3282 on the posterior probability for H_0 when $p=\frac{1}{4}$. Further examination of this example is possible, such as pairwise comparisons with the standard formulation and a composite null hypothesis, since bio-equivalence is often accepted if the ratio of a parameter in question for a new formulation to that for the standard lies between given limits.

4. Composite null hypothesis

A point null hypothesis is seldom exactly true and the real question an experimenter usually asks, is: "Is the parameter close enough to a specified value so as to make no real difference to the consequences of accepting the point null hypothesis?"

In this section we derive the posterior distribution of a binomial parameter when the null hypothesis is defined as an interval of length t around the point $\theta = \frac{1}{2}$.

So let

$$H_0: \theta \in \bar{\Theta}_0, \text{ where } \bar{\Theta}_0 = (\tfrac{1}{2}(1-t), \tfrac{1}{2}(1+t))$$

and

$$H_1: \theta \in \bar{\Theta}_1, \text{ where } \bar{\Theta}_1 \text{ is the complement of } \bar{\Theta}_0 \text{ in } \bar{\Theta} = (0,1).$$

Let the mass under the null hypothesis be distributed uniformly over $\bar{\Theta}_0$ while the mass under H_1 follows a symmetric Beta distribution over $\bar{\Theta}_1$.

The prior is then given by

$$(4.1) \quad \pi(\theta) = \begin{cases} \frac{p}{t} & , \theta \in \bar{\Theta}_0 \\ \frac{1-p}{1-\phi_t(\alpha+1, \alpha+1)} \frac{1}{B(\alpha+1, \alpha+1)} \theta^\alpha (1-\theta)^\alpha & , \theta \in \bar{\Theta}_1 \end{cases}$$

where

$$(4.2) \quad \begin{aligned} \phi_t(\alpha+1, \alpha+1) &= \int_{\bar{\Theta}_0} \frac{1}{B(\alpha+1, \alpha+1)} \theta^\alpha (1-\theta)^\alpha d\theta \\ &= I_{\frac{1}{2}(1+t)}(\alpha+1, \alpha+1) - I_{\frac{1}{2}(1-t)}(\alpha+1, \alpha+1) \end{aligned}$$

and $I_k(a, b) = \frac{B_k(a, b)}{B(a, b)}$ where $B_k(a, b)$ is the incomplete Beta function.

The posterior density function is then given by

$$(4.3) \quad \pi_0(\theta|x) = \psi(t, \alpha) \frac{1}{B(x+1, n-x+1)} \theta^x (1-\theta)^{n-x}, \quad \theta \in \bar{\Theta}_0$$

$$(4.4) \quad \pi_1(\theta|x) = \psi(t, \alpha) \frac{(1-p)t\theta^{x+\alpha}(1-\theta)^{n-x+\alpha}}{p[1-\phi_t(\alpha+1, \alpha+1)]B(\alpha+1, \alpha+1)B(x+1, n-x+1)},$$

$\theta \in \bar{\Theta}_1$

where

$$(4.5) \quad \psi(t, \alpha) = \left[\phi_t(x+1, n-x+1) + \frac{(1-p)t^{1-\phi_t(x+\alpha+1, n-x+\alpha-1)}B(x+\alpha+1, n-x+\alpha+1)}{p[1-\phi_t(\alpha+1, \alpha+1)]B(\alpha+1, \alpha+1)B(x+1, n-x+1)} \right]^{-1}.$$

From (4.3) and (4.4) follow the posterior probabilities

$$(4.6) \quad P[\theta \in \bar{\Theta}_0|x] = \psi(t, \alpha)\phi_t(x+1, n-x+1),$$

$$(4.7) \quad P[\theta > \frac{1}{2}(1+t)|x] = \psi(t, \alpha) \frac{(1-p)tB(x+\alpha+1, n-x+\alpha+1)[1-I_{\frac{1}{2}}(1+t)(x+\alpha+1, n-x+\alpha+1)]}{p[1-\phi_t(\alpha+1, \alpha+1)]B(\alpha+1, \alpha+1)B(x+1, n-x+1)}$$

$$(4.8) \quad P[\theta < \frac{1}{2}(1-t)|x] = \psi(t, \alpha) \frac{(1-p)tB(x+\alpha+1, n-x+\alpha+1)I_{\frac{1}{2}}(1-t)(x+\alpha+1, n-x+\alpha+1)}{p[1-\phi_t(\alpha+1, \alpha+1)]B(\alpha+1, \alpha+1)B(x+1, n-x+1)}.$$

When $t \rightarrow 0$ the posterior probabilities (4.6) - (4.8) reduce to those given in (2.4) and following from (2.5) for $k=2$ (see appendix B). These posterior probabilities turn out to be remarkably robust with respect to moderate changes in t and α . To illustrate this, $P[\theta \in \bar{\Theta}_0|x=9]$ is tabulated for various values of t and α when $n=12$.

Table 2. $P[\theta \in \Theta_0 | x=9]$ where $n=12$, $p=\frac{1}{2}$

α	t	0	0,5	0,10	0,2
0		0,4112	0,4099	0,4133	0,4339
1		0,3793	0,3745	0,3738	0,3847
4		0,3868	0,3775	0,3713	0,3688

So we can see that a moderate (and in some cases not so moderate) interval around $\theta = \frac{1}{2}$ under H_0 changes the posterior probability very little, so even if the null hypothesis is acceptable when θ is reasonably close to $\frac{1}{2}$, a point mass under H_0 will usually be sufficient. It is interesting to note that there seems to exist a lower bound on the posterior probability for H_0 with respect to t for fixed α and x . The posterior probability is not necessarily a monotonic increasing function of the length of the interval. By defining the null hypothesis as an interval, the mass p is diffused and it can lead to a decrease of posterior probability if the likelihood does not support H_0 . For example, in table 2 for $\alpha=0$, the minimum posterior probability of 0,4098 occurs at $t=0,04$ while for $\alpha=1,7$ we get a minimum of 0,3675 at $t=0,11$.

5. Loss function

In a decision-making situation the final decision will depend heavily on the possible consequences of a wrong decision and the natural Bayesian way to proceed would be to define the loss functions $L_0(\theta)$ and $L_1(\theta)$ where, say,

$$(5.1) \quad L_0(\theta) = \sum_{i=1}^k \left(\theta_i - \frac{1}{k} \right)^2$$

is the loss incurred when H_0 is accepted, while

$$(5.2) \quad L_1(\theta) = \begin{cases} b & , \quad \theta_i = \frac{1}{k} \quad , \quad i=1, \dots, k. \\ 0 & \text{otherwise} \end{cases}$$

is the loss when H_1 is accepted.

The posterior expected loss is then given by ρ_i , $i=0,1$, for accepting H_i , where

$$(5.3) \quad \rho_i = E[L_i(\theta) | \mathbf{x}]$$

and the expectation is over the posterior distribution of θ . The Bayes decision is then to accept H_0 if $\rho_0 < \rho_1$ and to accept H_1 otherwise. It follows that

$$(5.4) \quad \rho_1 = b\pi_0(\theta | \mathbf{x})$$

and

$$(5.5) \quad \rho_0 = \int_{\Theta_1} \sum_{i=1}^k \left(\theta_i - \frac{1}{k} \right)^2 \pi_1(\theta | \mathbf{x}) d\theta$$

where $\pi_i(\theta | \mathbf{x})$, $i=0,1$, are given in (2.4) and (2.5).

Equation (5.5) reduces to

$$(5.6) \quad \rho_0 = \pi_0(\theta | \mathbf{x})^{\frac{1-p}{p}} \frac{k^n \Gamma[k(\alpha+1)]}{(\Gamma[\alpha+1])^k} \left[\sum_{i=1}^k \frac{\Gamma(x_i + \alpha + 1) \prod_{j \neq i} \Gamma(x_j + \alpha + 1)}{\Gamma[n+k(\alpha+1)+2]} - \frac{1}{k} \frac{\Gamma[x_i + \alpha + 1]}{\Gamma[n+k(\alpha+1)]} \right].$$

Now ρ_0 and ρ_1 are functions of α and the decision to accept H_0 will be robust with respect to α if $\rho_0 < \rho_1$ for all α . On the other hand, ρ_0 can never be larger than ρ_1 for all α , since $\rho_0 \rightarrow 0$ when $\alpha \rightarrow \infty$ while ρ_1

is a convex function of α with positive lower bound.

If the decision-maker feels that the losses suffered under both kinds of wrong decisions are comparable in magnitude or, in other words, that the loss suffered when H_1 is wrongly accepted is approximately equal to the expected loss when H_0 is wrongly accepted, then let

$$(5.7) \quad b = E \left[\sum_{i=1}^k \left(\theta_i - \frac{1}{k} \right)^2 \right] \\ = \frac{k-1}{k(k\alpha+k+1)},$$

where the expectation is over the prior distribution of θ .

To illustrate the behaviour of the risk functions, let us consider the hypothetical data (8,15,4) from a trinomial distribution with loss functions as in (5.1) and (5.2) with $k=3$. From (2.4) with uniform prior it follows that

$$P \left[\theta = \frac{1}{3} e \mid x = (8,15,4) \right] = 0,3132$$

with a lower bound of 0,2230 at $\alpha = 2,5$.

In figure 2 ρ_0 and ρ_1 are shown as functions of α and for two values of b .

From figure 2 it follows that $\rho_0 < \rho_1$ for all α when $b = 0,24$. So the Bayes decision to accept the null hypothesis would be robust with respect to the exact form of the prior when $b \geq 0,24$. The value of b at which the decision to accept the alternative hypothesis would be robust, is not so clear cut. We feel that to have a robust decision in favour of H_1 , ρ_1 must be smaller than ρ_0 for a reasonable range of priors, including the uniform prior ($\alpha=0$). For $b = 0,19$ it follows that $\rho_1 < \rho_0$ for $-0,02 < \alpha < 3,16$ which would lead us to accept H_1 for $b \leq 0,19$. While the decision will be clear cut for values of $b \geq 0,24$ or $b \leq 0,19$, these decisions could still be acceptable for values of b between them, but close

to either one. So for this example our decision would be to:

accept H_0 if $b \geq 0,23$;

accept H_1 if $b \leq 0,20$.

For values of b in the range $0,20 - 0,23$, we are in the sensitivity zone and a re-examination of the parameters and possibly more data are recommended.

If b is defined by (5.7), then $\rho_1 < \rho_0$ for $\alpha > 0,1$, which would lead us to accept the alternative hypothesis.

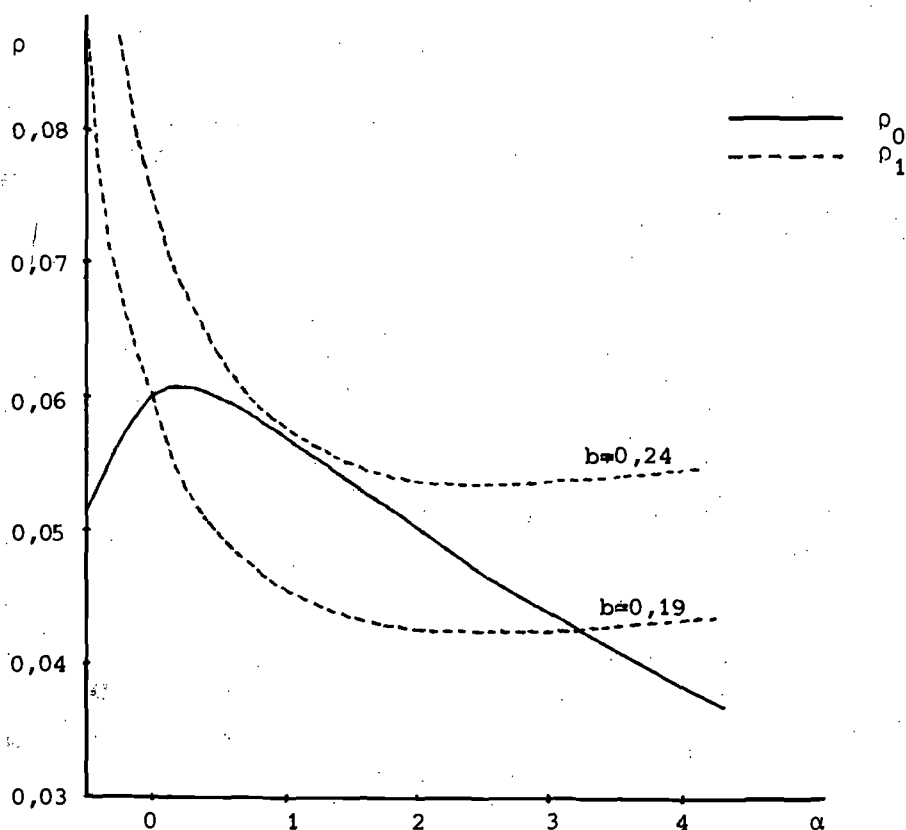


Figure 2. Posterior expected loss

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Appendix A

For completeness the boundaries of integration are given here for the probabilities stated in (3.6).

$$P[A \text{ is best} | \mathbf{x} = (2, 1, 7, 2)] \\ = \int \int \int S^{\pi_1(\theta | \mathbf{x})} d\theta_A d\theta_F d\theta_L$$

where $\theta_P = 1 - \theta_A - \theta_F - \theta_L$ and $\pi_1(\theta | \mathbf{x})$ is given in (2.5) with $\alpha = 0$. S is the subspace of the parameter space where θ_A is larger than the other three θ 's.

$$S = \{(\theta_A, \theta_F, \theta_L):$$

$$\begin{aligned} &0 < \theta_L < 1 - \theta_F - \theta_A, \quad 0 < \theta_F < 1 - \theta_A, \quad \frac{1}{2} < \theta_A < 1; \\ &1 - 2\theta_A - \theta_F < \theta_L < \theta_A, \quad 0 < \theta_F < 1 - 2\theta_A, \quad \frac{1}{3} < \theta_A < \frac{1}{2}; \\ &0 < \theta_L < 1 - \theta_F - \theta_A, \quad 1 - 2\theta_A < \theta_F < \theta_A, \quad \frac{1}{3} < \theta_A < \frac{1}{2}; \\ &1 - 2\theta_A - \theta_F < \theta_L < \theta_A, \quad 1 - 3\theta_A < \theta_F < \theta_A, \quad \frac{1}{4} < \theta_A < \frac{1}{3} \}. \end{aligned}$$

Appendix B

From (4.5) and (4.6) it follows that

$$P[\theta \in \Theta_0 | \mathbf{x}] = \left[1 + \frac{1-p}{p} \frac{t}{\phi_t(x+1, n-x+1)} \frac{1 - \phi_t(x+\alpha+1, n-x+\alpha+1)}{1 - \phi_t(\alpha+1, \alpha+1)} \right]^{-1} \cdot \frac{B(x+\alpha+1, n-x+\alpha+1)}{B(\alpha+1, \alpha+1) B(x+1, n-x+1)}$$

Since

$$\frac{1-\phi_t(x+\alpha+1, n-x+\alpha+1)}{1-\phi_t(\alpha+1, \alpha+1)} \rightarrow 1 \text{ as } t \rightarrow 0,$$

we will examine the limiting behaviour of

$$\frac{1}{t} \phi_t(x+1, n-x+1) B(x+1, n-x+1) = \gamma(t), \text{ say.}$$

Now

$$\begin{aligned} \gamma(t) &= \frac{1}{t} \int_{\frac{1}{2}(1-t)}^{\frac{1}{2}(1+t)} \theta^x (1-\theta)^{n-x} d\theta \\ &= \frac{1}{t} \sum_{k=0}^{n-x} (-1)^k \binom{n-x}{k} \int_{\frac{1}{2}(1-t)}^{\frac{1}{2}(1+t)} \theta^{x+k} d\theta \\ &= \frac{1}{t} \sum_{k=0}^{n-x} \frac{(-1)^k}{x+k+1} \binom{n-x}{k} \left(\frac{1}{2}\right)^{x+k+1} \left[(1+t)^{x+k+1} - (1-t)^{x+k+1} \right] \\ &= \frac{1}{t} \sum_{k=0}^{n-x} \frac{(-1)^k}{x+k+1} \binom{n-x}{k} \left(\frac{1}{2}\right)^{x+k+1} \left[2 \sum_{j=1}^m \binom{x+k+1}{2j-1} t^{2j-1} \right] \end{aligned}$$

where

$$m = \begin{cases} \frac{x+k}{2} + 1 & \text{for } x+k \text{ even} \\ \frac{k+x-1}{2} + 1 & \text{for } x+k \text{ odd.} \end{cases}$$

So

$$\gamma(t) = \sum_{k=0}^{n-x} \frac{(-1)^k}{x+k+1} \binom{n-x}{k} \left(\frac{1}{2}\right)^{x+k} \left[\sum_{j=1}^m \binom{x+k+1}{2j-1} t^{2(j-1)} \right],$$

and

$$\begin{aligned} \lim_{t \rightarrow 0} \gamma(t) &= \sum_{k=0}^{n-x} (-1)^k \binom{n-x}{k} \left(\frac{1}{2}\right)^{x+k} \\ &= \left(\frac{1}{2}\right)^n. \end{aligned}$$

This means that

$$\lim_{t \rightarrow 0} P[\theta \in \Theta_0 | x] = \left[1 + \frac{1-p}{p} \frac{2^n B(x+\alpha+1, n-x+\alpha+1)}{B(\alpha+1, \alpha+1) B(x+1, n-x+\alpha+1)} \right]^{-1}$$

which is the same as (2.4) when $k=2$.

Using the same limiting result it follows easily that (4.7) and (4.8) reduce to the posterior probabilities obtainable from (2.5) if $k=2$.

References

- BERGER, J.O. (1985). *Statistical decision theory*. 2nd ed. Springer Verlag, New York.
- BERGER, J.O. (1986). Personal communication with authors.
- CASELLA, G. & R.L. BERGER, (1987). Reconciling Bayesian and Frequentist Evidence in the One-sided Testing Problem. *Journal of the Am. Statist. Ass.* Vol. 82, No. 397, 106-111.
- DEGROOT, M.H. (1958). *Optimal statistical decisions*. McGraw-Hill, New York.
- DICKEY, J.M. (1977). Is the tail area useful as an approximate Bayes Factor? *Journal of the Am. Statist. Ass.*, Vol. 72, No. 357, 138-142.
- GOOD, I.J. (1967). A Bayesian significance test for multinomial distributions (with discussion). *J. Royal Statist. Soc. B.* 29, 399-431.
- JEFFREYS, H. (1961). *Theory of probability*. 3rd ed. Oxford University Press.
- MEYER, B.H., MULLER, F.O., SWART, K.J., LUUS, H.G. & WERKMAN, I.M. (1985). Comparative bio-availability of four formulations of Furosimide. *S.A. Med. J.* 68, 645-647.
- PRATT, J.W. (1965). Bayesian interpretation of standard inference statements (with discussion). *J. Royal Statist. Soc. B.* 27, 169-203.